Appendix A
Simulation Pseudocode for an ABM of Brokerage in Matching Markets

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, or matching rules.

For any matrix Z, $Z_{\cdot\ell}$ denotes column ℓ and $Z_{\cdot L}$ denotes the submatrix with ordered column-index set L. For a vector z, z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
Core		
N	scalar	Number of principals, with principal set $[N] = \{1, \ldots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t.
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \ldots; m^T]$	$\mathbb{R}^{T \times m}$	Measure table; row $m^t = \mathcal{M}_{t\cdot} \in \mathbb{R}^m$.
Parameters		
d	scalar	Principal type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction update weight (exponentially weighted moving average, EWMA).
L	learning function	Neural network (NN, default) or ridge regression.
S_b	broker service	Full (default), assessment-only, or access-only.
λ_a, λ_b	positive scalars	Principal and broker ridge slope penalties; calibrated experiment values are $\lambda_a = 0.001$ and $\lambda_b = 0.001$.
a_b	broker ridge variant	Pair, size-matched, single-principal, or additive.
$E_{\text{init},a}, E_{\text{init},b}$	scalars	Principal and broker initial training steps; both are 100 in the reporting simulations.
$E_{\text{steps},a}, E_{\text{steps},b}$	scalars	Principal and broker minimum training steps per fitting call; both are 50 in the reporting simulations.
$\eta_{\text{lr},a}, \eta_{\text{lr},b}$	scalars	Principal and broker Adam learning rates; reporting values are 0.003 and 0.03, respectively.
W_p	scalar	Training-window horizon, in periods.
M_{train}	scalar	Per-call training observation cap.

Symbol	Dimension/domain	Definition
Principal types, networks, and output		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N+1\}$; node $N+1$ is the broker.
$X^t = [x_1^t, \ldots, x_N^t]$	$\mathbb{R}^{d \times N}$	Principal type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0,1] \to \mathbb{R}^d$	Principal type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized match output for pair (i,j).
Match-output function		
Q	scalar	Additive offset in underlying match value.
ρ	scalar	Controls the relative weighting of individual quality and complementarity: $\rho = 0$ gives pure complementarity; $\rho = 1$ gives pure individual quality.
δ	scalar	Strength of the pair boundary in match-specific complementarity.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Common ideal principal type.
A	$\mathbb{R}^{d \times d}$	Symmetric positive definite complementarity matrix.
B	$\mathbb{R}^{d \times d}$	Symmetric boundary matrix.
Calibration and costs		
$\bar{q}_{\text{cal}}$	scalar	Calibration mean.
r	scalar	Reservation threshold.
ϕ	scalar	Broker fee.
c_s	scalar	Self-search cost.
Learning and predictions		
d_B	integer	Broker symmetric pair-feature dimension, $d + d(d+1)/2$.
$\psi_b(x_i, x_j)$	$\mathbb{R}^{d_B}$	Symmetric broker pair feature vector.
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d \times n_i^t}$, $y_i^t \in \mathbb{R}^{n_i^t}$	Principal i history; observation ℓ is partner principal type $\Xi_{i,\cdot\ell}^t$ and realized match output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (X_{b,1}^t, X_{b,2}^t, y_b^t, n_b^t)$	$X_{b,1}^t, X_{b,2}^t \in \mathbb{R}^{d \times n_b^t}$, $y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ stores the two party types and realized match output $y_{b\ell}^t$.
Θ_i^t	predictor parameters	Principal i NN weights or ridge coefficients.
Θ_b^t	predictor parameters	Broker NN weights or ridge coefficients.
$\mathcal{O}_i^t, \mathcal{O}_b^t$	Adam states	Persistent NN optimizer moments and step counters; unused by ridge.
Δ_i^t, Δ_b^t	nonnegative integers	Observations added since the learner's previous training call.
$\mathbf{p}_i^t, \mathbf{p}_b^t$	integer sequences	Cumulative history counts at the end of learner initialization (period 0) and each later completed period.
$v_b^t(i,j)$	scalar	Broker's predicted match value for unordered pair $\{i,j\}$.
$a_{i \to j}^t$	scalar	Principal i's assessment of candidate j.
$w_{j \leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_{i\cdot}^t$	vector	Principal i's mean realized match output by partner.
$C_{i\cdot}^t$	vector	Principal i's partner-indexed counts.
Period state and market objects		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that principal i has ever used the broker.

Symbol	Dimension/domain	Definition
M_i^t	set	Principal i current-period counterparties.
z_i^t	channel	Channel chosen by demander i.
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records $(i, j, z, q, \hat{q}, o_1, o_2)$.
rng^t	RNG state	State of the simulation random-number stream.

The mutable simulation state at the end of period t is

$$S^t = \begin{pmatrix} \vartheta, \mathsf{rng}^t, \gamma, c, A, B, \bar{q}_{\text{cal}}, r, \phi, c_s, X^t, G^t, \\ \{\mathcal{H}_i^t, \Theta_i^t, \mathcal{O}_i^t, \Delta_i^t, \mathbf{p}_i^t, \bar{q}_{i\cdot}^t, C_{i\cdot}^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \\ \mathcal{H}_b^t, \Theta_b^t, \mathcal{O}_b^t, \Delta_b^t, \mathbf{p}_b^t, \text{Roster}^t, D^t, \text{rep}_b^t \end{pmatrix}.$$

This tuple includes fixed realized model objects and all variables needed for later transitions. Implementation-only scratch buffers and diagnostic-only counters are omitted.

The fixed calibration moments are calculated across all unordered pairs of distinct principals within each of 50 independently generated calibration markets of 1,000 principals, then averaged with equal weight across markets. Define

$$U_{ij} = \frac{x_i^\top c + x_j^\top c}{2}, \qquad V_{ij} = x_i^\top A x_j, \qquad H_{ij} = \text{sign}(x_i^\top B x_j) V_{ij},$$

$$\widetilde{U}_{ij} = \frac{U_{ij} - \mu_U}{\sigma_U}, \qquad \widetilde{W}_{ij}(\delta) = \frac{V_{ij} + \delta H_{ij} - (\mu_V + \delta \mu_H)}{\sqrt{\sigma_V^2 + 2\delta \,\text{Cov}(V, H) + \delta^2 \sigma_H^2}}.$$

With $\kappa(\delta)$ denoting the average within-market covariance between the standardized individual-quality and complementarity components in the calibration markets, the underlying match value excluding Q is

$$f(x_i, x_j) = \frac{\sqrt{\rho}\,\widetilde{U}_{ij} + \sqrt{1-\rho}\,\widetilde{W}_{ij}(\delta)}{\sqrt{1 + 2\sqrt{\rho(1-\rho)}\,\kappa(\delta)}}.$$

The normalization coefficients depend on ρ and δ, are shared across simulations, and remain fixed through turnover. For a given pair, underlying match value is $Q + f(x_i, x_j)$. A completed match produces realized match output

$$q_{ij} = Q + f(x_i, x_j) + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

2 Top-Level Simulation

Algorithm 1 Brokerage Simulation

Require: Parameters ϑ and seed.
Ensure: Final state S^T and measure table $\mathcal{M} \in \mathbb{R}^{T \times m}$.

$S^0 \leftarrow \textsc{Initialize}(\vartheta, \text{seed})$
for $t = 1, \ldots, T$ **do**
    $(S^t, \mathcal{R}^t, Y^t) \leftarrow \textsc{PeriodUpdate}(S^{t-1}, t, \vartheta)$
    $\mathcal{M}_{t\cdot} \leftarrow Y^t$
return $(S^T, \mathcal{M})$

3 Learning and Prediction

Let $\mathsf{L} \in \{\text{NN}, \text{Ridge}\}$ select the learning function. The default is $\mathsf{L} = \text{NN}$. Both branches use the same histories, period window, and observation cap. Inputs remain on their original scale.

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h,

$$\text{NN}_\Theta(\xi) = w_2^\top \,\text{ReLU}(W_1\xi + b_1) + b_2, \qquad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h \times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$.

$$d_B = d + \frac{d(d+1)}{2}, \qquad \psi_b(x_i, x_j) = \left[x_i + x_j;\ \text{vech}\left(\frac{x_i x_j^\top + x_j x_i^\top}{2} \right) \right].$$

Principal networks use $(p_{\text{in}}, h) = (d, 2d)$; the broker network uses $(p_{\text{in}}, h) = (d_B, 8d)$.

$$\psi_b(x_i, x_j) = \psi_b(x_j, x_i)$$

so broker histories produce one training column per observed unordered pair.

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
L	learning function	NN or Ridge.
a_b	broker ridge variant	Pair, size-matched, single-principal, or additive.
Θ	predictor parameters	NN parameters (W_1, b_1, w_2, b_2) or ridge parameters (α, β).
W_1, b_1, w_2, b_2	matrix, vectors, scalar	Neural network weights and biases.
d_B	integer	Broker symmetric pair-feature dimension.
ψ_b	function	Symmetric broker pair-feature map.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}} \times n}$, $\mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch mean squared error (MSE) objective.
λ	positive scalar	Learner-specific ridge slope penalty; intercepts are not penalized.
e	integer	Update-step index.
(m, v, τ)	Adam state	Persistent Adam moments and step counter; $\mathcal{O}_i, \mathcal{O}_b$ are the principal and broker instances.
$\beta_1, \beta_2, \epsilon$	scalars	Adam constants (0.9, 0.999, 10^{-8}).
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t.

Local variables for training (continued)

Symbol	Dimension/domain	Definition
L_i, L_b	ordered index sets	Training-window indices from the most recent W_p completed simulation periods, with period 0 included only until W_p later periods exist; subsampled evenly to at most M_{train}.
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Principal and broker NN update-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d\times\lvert L_i\rvert}$, $\mathbb{R}^{\lvert L_i\rvert}$	Principal i training view of $\mathcal{H}_i^t$.
$Z_{b,\text{tr}}, y_{b,\text{tr}}$	matrix, $\mathbb{R}^{\lvert L_b\rvert}$	Broker fitting view using the selected input map.
r_ℓ	$\{1,2\}$	Endpoint retained once when broker observation ℓ is recorded for the single-principal ablation.

Algorithm 2 TrainNN

Require: Network parameters Θ, inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}}\times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E, learning rate η_{lr}, persistent Adam state (m, v, τ).

for $e = 1, \dots, E$ **do**
  $g \leftarrow \nabla_\Theta \mathcal{L}(\Theta), \quad \mathcal{L}(\Theta) = n^{-1}\sum_{\ell=1}^{n}\left(\text{NN}_\Theta(\Xi_{\text{tr},\cdot\ell}) - y_{\text{tr},\ell}\right)^2$
  $\tau \leftarrow \tau + 1$
  $m \leftarrow \beta_1 m + (1-\beta_1)g, \quad v \leftarrow \beta_2 v + (1-\beta_2)\, g \odot g$
  $\Theta \leftarrow \Theta - \eta_{\text{lr}} \dfrac{m/(1-\beta_1^\tau)}{\sqrt{v/(1-\beta_2^\tau)} + \epsilon}$
return $\Theta, (m, v, \tau)$

Algorithm 3 FitPredictor

Require: Learning function L, learner $c \in \{a, b\}$, parameters Θ, raw inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}}\times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, learner-specific ridge penalty λ_c, and fitting parameters.

if $\mathsf{L} = \text{NN}$ **then**
  $E \leftarrow \max\{E_{\text{steps},c}, \lceil E_{\text{init},c}\Delta/n_{\text{total}}\rceil\}$.
  $(\Theta, \mathcal{O}) \leftarrow \textsc{TrainNN}(\Theta, \Xi_{\text{tr}}, y_{\text{tr}}, E, \eta_{\text{lr},c}, \mathcal{O})$.
else
  $(\alpha, \beta) \leftarrow \arg\min_{a,b}\left\{ n^{-1}\sum_{\ell=1}^{n}(a + b^\top \Xi_{\text{tr},\cdot\ell} - y_{\text{tr},\ell})^2 + \lambda_c\|b\|_2^2 \right\}$.
  $\Theta \leftarrow (\alpha, \beta)$.
return Θ and, for NN, $\mathcal{O}$.

The principal input is always the prospective partner type. The broker ridge training input for observation ℓ is

$$z_{b\ell} = \begin{cases} \psi_b(x_{1\ell}, x_{2\ell}), & a_b \in \{\text{pair}, \text{size-matched}\}, \\ x_{r_\ell,\ell}, & a_b = \text{single-principal}, \\ x_{1\ell} + x_{2\ell}, & a_b = \text{additive}. \end{cases}$$

For size-matched history, uniformly sample without replacement from the broker's windowed and capped observations. Its sample size is the smaller of the broker sample size and the nearest-integer median principal sample size among principals with nonempty histories, with half-integers rounded up.

Prediction is modular:

$$\textsc{PredictPrincipal}(i, x_j) = \begin{cases} \mathrm{NN}_{\Theta_i}(x_j), & \mathsf{L} = \mathrm{NN}, \\ \alpha_i + \beta_i^\top x_j, & \mathsf{L} = \mathrm{Ridge}, \end{cases}$$

and

$$\textsc{PredictBroker}(i, j) = \begin{cases} \mathrm{NN}_{\Theta_b}(\psi_b(x_i, x_j)), & \mathsf{L} = \mathrm{NN}, \\ \alpha_b + \gamma_b^\top \psi_b(x_i, x_j), & \mathsf{L} = \mathrm{Ridge},\ a_b \in \{\text{pair}, \text{size-matched}\}, \\ \alpha_b + \gamma_b^\top (x_i + x_j), & \mathsf{L} = \mathrm{Ridge},\ a_b = \text{additive}, \\ g_b(x_i) + g_b(x_j) - \bar{y}_{b,\mathrm{tr}}, & \mathsf{L} = \mathrm{Ridge},\ a_b = \text{single-principal}, \end{cases}$$

where $g_b(x) = \alpha_b + \gamma_b^\top x$. Before any ridge fit, $\alpha = Q$, slopes are zero, and $\bar{y}_{b,\mathrm{tr}} = Q$.

Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 4 TrainPredictors

Require: State S, period t, learning function L, and fitting parameters.

for $i = 1, \ldots, N$ in parallel **do**
  $n_i \leftarrow n_i^t$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.
  if $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**
    Let $P_i \leftarrow |\mathbf{p}_i|$ and $b_i \leftarrow 0$ if $P_i \le W_p$, otherwise $b_i \leftarrow (\mathbf{p}_i)_{P_i - W_p}$; set $L_i \leftarrow \{b_i + 1, \ldots, n_i\}$. If $|L_i| > M_{\mathrm{train}}$, keep M_{train} spaced indices across L_i, including its first and last indices (or only its last index when $M_{\mathrm{train}} = 1$).
    $\Xi_{i,\mathrm{tr}} \leftarrow (\Xi_i^t)_{\cdot L_i}$, $y_{i,\mathrm{tr}} \leftarrow (y_i^t)_{L_i}$.
    $\Theta_i \leftarrow \textsc{FitPredictor}(\mathsf{L}, a, \Theta_i, \Xi_{i,\mathrm{tr}}, y_{i,\mathrm{tr}}, \lambda_a, \Delta_i, n_i, \vartheta)$; $\Delta_i \leftarrow 0$.
$n_b \leftarrow n_b^t$, $\Delta_b \leftarrow$ broker observations since last training.
if $n_b > 0$ and $\Delta_b > 0$ **then**
  Let $P_b \leftarrow |\mathbf{p}_b|$ and $b_b \leftarrow 0$ if $P_b \le W_p$, otherwise $b_b \leftarrow (\mathbf{p}_b)_{P_b - W_p}$; set $L_b \leftarrow \{b_b + 1, \ldots, n_b\}$. If $|L_b| > M_{\mathrm{train}}$, keep M_{train} spaced indices across L_b, including its first and last indices (or only its last index when $M_{\mathrm{train}} = 1$).
  If $\mathsf{L} = \mathrm{Ridge}$ and $a_b =$ size-matched, replace L_b by the uniform size-matched sample defined above.
  Construct $Z_{b,\mathrm{tr}} = [z_{b\ell}]_{\ell \in L_b}$ and $y_{b,\mathrm{tr}} \leftarrow (y_b^t)_{L_b}$.
  $\Theta_b \leftarrow \textsc{FitPredictor}(\mathsf{L}, b, \Theta_b, Z_{b,\mathrm{tr}}, y_{b,\mathrm{tr}}, \lambda_b, \Delta_b, n_b, \vartheta)$; $\Delta_b \leftarrow 0$.

4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\mathrm{cal}} = \frac{1}{10{,}000} \sum_{\ell=1}^{10{,}000} \{Q + f(x_{i_\ell}, x_{j_\ell})\},$$
$$r = f_r\, \bar{q}_{\mathrm{cal}}, \qquad \phi = c_s = \lambda_c\, \bar{q}_{\mathrm{cal}}.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s.
f_r	scalar	Reservation coefficient; sets $r = f_r\,\bar{q}_{\mathrm{cal}}$ (default 0.60).
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ.
ψ_ℓ	scalar	Phase for active curve coordinate ℓ.
A_0	$\mathbb{R}^{d\times d}$	Raw matrix used to construct A.
Σ_x	$\mathbb{R}^{d\times d}$	Empirical principal type second moment.
B_0	$\mathbb{R}^{d\times d}$	Raw zero-trace symmetric boundary matrix.
B_{raw}	$\mathbb{R}^{d\times d}$	Orthogonalized pre-normalization boundary matrix.
$Z_{b,\mathrm{seed}}, y_{b,\mathrm{seed}}$	matrix, $\mathbb{R}^{n_b^0}$	Broker seed fitting batch using the selected input map.

Local variables for InitializePrincipal

Symbol	Dimension/domain	Definition
σ_x	scalar	Principal type noise scale.
u_i	scalar	Principal i curve position.
ϵ_i	$\mathbb{R}^d$	Principal type noise for principal i.
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\mathrm{ent}}$	subset of $[N]\setminus\{i\}$	Entrant attachment set.

Algorithm 5 InitializePrincipal

Require: State S, principal slot i, period counter t, parameters ϑ.
Ensure: Principal slot i is initialized in S.

Draw $u_i \sim \mathrm{Unif}[0,1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
$x_i^t \leftarrow (\gamma(u_i) + \epsilon_i)/\|\gamma(u_i) + \epsilon_i\|_2$.
Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_{i\cdot}^t$, $C_{i\cdot}^t$, χ_i^t, M_i^t, and Δ_i^t.
Initialize the selected predictor to the constant Q: for NN, draw W_{1i}^t with He initialization and set $b_{1i}^t = 0$, $w_{2i}^t = 0$, $b_{2i}^t = Q$, $\mathcal{O}_i^t = 0$; for ridge, set $\alpha_i^t = Q$ and $\beta_i^t = 0$.
if $t = 0$ **then**
    $s_i^t \leftarrow (0, 0)$.
else
    Remove i from Roster^t, D^t, and M_j^t for all j; delete all incident edges in G^t; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.
    Sample $\mathcal{V}_i^{\mathrm{ent}} \subset [N]\setminus\{i\}$ without replacement, $|\mathcal{V}_i^{\mathrm{ent}}| = \min(\lfloor k_G/2 \rfloor, N-1)$, with

$$\pi_{ij}^{\mathrm{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$

    Add edges (i,j) to G^t for all $j \in \mathcal{V}_i^{\mathrm{ent}}$.
    $s_{i,\mathrm{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\mathrm{ent}}|^{-1} \sum_{j\in\mathcal{V}_i^{\mathrm{ent}}} s_{j,\mathrm{self}}^t, & \mathcal{V}_i^{\mathrm{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\mathrm{ent}} = \emptyset. \end{cases}$
    $s_{i,\mathrm{broker}}^t \leftarrow \mathrm{rep}_b^t$.
    Set $\mathbf{p}_i^t \leftarrow (0)$, treating entrant initialization as period 0 of its tenure.

Algorithm 6 Initialize

Require: Parameters ϑ and seed.
Ensure: Initial state S^0.

Draw curve frequencies $\nu_\ell \sim \mathrm{Unif}\{1,\dots,5\}$ and phases $\psi_\ell \sim \mathrm{Unif}[0,2\pi)$ for $\ell = 1,\dots,d_\gamma$, defining γ.
Allocate empty principal and broker containers in S^0 and store γ.
for $i = 1,\dots,N$ **do**
    InitializePrincipal$(S^0, i, 0, \vartheta)$.
Draw $u_c \sim \mathrm{Unif}[0,1]$ and $\epsilon_c \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$; set $c \leftarrow [\gamma(u_c) + \epsilon_c]/\|\gamma(u_c) + \epsilon_c\|_2$.
Draw $A_0 \in \mathbb{R}^{d\times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d/\operatorname{tr}(A_0^\top A_0)$.
$\Sigma_x \leftarrow N^{-1}\sum_i x_i^0 (x_i^0)^\top$.
Draw symmetric zero-trace B_0 and set

$$B_{\mathrm{raw}} \leftarrow B_0 - \frac{\operatorname{tr}(\Sigma_x B_0 \Sigma_x A)}{\operatorname{tr}(\Sigma_x A \Sigma_x A)} A, \qquad B \leftarrow B_{\mathrm{raw}}/\|B_{\mathrm{raw}}\|_F.$$

Calibrate $(\bar{q}_{\mathrm{cal}}, r, \phi, c_s)$.
Build G^0 as a Watts-Strogatz graph on $[N]$ with degree k_G and rewiring probability p_{rewire}; add isolated broker node $N+1$.
Initialize $\mathcal{H}_b^0$ and the selected broker predictor to the constant Q using the corresponding NN or ridge initialization above; set $\mathrm{rep}_b^0 = 0$.
Sample $\mathrm{Roster}^0 \subset [N]$ uniformly with $|\mathrm{Roster}^0| = \lceil \alpha_R N \rceil$; add broker edges $(i, N+1)$ for all roster members.
for each unordered non-broker edge $\{i,j\} \in \mathcal{E}_G^0$ **do**
    Draw q_{ij} once; append (x_j^0, q_{ij}) to $\mathcal{H}_i^0$ and (x_i^0, q_{ij}) to $\mathcal{H}_j^0$; update $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$.
Let $\mathcal{B}_{\mathrm{seed}} \leftarrow \{\{i,j\} \in \mathcal{E}_G^0 : i,j \in \mathrm{Roster}^0\}$; shuffle and keep at most 100 edges.
for each $\{i,j\} \in \mathcal{B}_{\mathrm{seed}}$ **do**
    Append (x_i^0, x_j^0, q_{ij}) to $\mathcal{H}_b^0$; increment n_b^0.
    If $\mathsf{L} =$ Ridge and $a_b =$ single-principal, draw and store $r_\ell \sim \mathrm{Unif}\{1,2\}$ for this observation.
if $n_b^0 > 0$ **then**
    Set rep_b^0 to mean broker seed output.
else
    Set $\mathrm{rep}_b^0 \leftarrow 0$.
Set $s_{i,\mathrm{self}}^0$ to mean own seed output and $s_{i,\mathrm{broker}}^0 \leftarrow \mathrm{rep}_b^0$.
Set $\mathbf{p}_i^0 \leftarrow (n_i^0)$ for every principal and $\mathbf{p}_b^0 \leftarrow (n_b^0)$, so initialization is period 0; set $\Delta_i^0 \leftarrow n_i^0$ and $\Delta_b^0 \leftarrow n_b^0$.
for each i with $n_i^0 > 0$ **do**
    $\Theta_i^0 \leftarrow$ FitPredictor$(\mathsf{L}, a, \Theta_i^0, \Xi_i^0, y_i^0, \lambda_a, n_i^0, n_i^0, \vartheta)$; $\Delta_i^0 \leftarrow 0$.
if $n_b^0 > 0$ **then**
    $Z_{b,\mathrm{seed}} \leftarrow [z_{b\ell}]_{\ell=1}^{n_b^0}$, $y_{b,\mathrm{seed}} \leftarrow y_b^0$.
    $\Theta_b^0 \leftarrow$ FitPredictor$(\mathsf{L}, b, \Theta_b^0, Z_{b,\mathrm{seed}}, y_{b,\mathrm{seed}}, \lambda_b, n_b^0, n_b^0, \vartheta)$; $\Delta_b^0 \leftarrow 0$.
return S^0.

5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 7 ChooseChannel

Require: Principal i state (s_i, χ_i), broker reputation rep_b, RNG.

$U_{\text{self}} \leftarrow s_{i,\text{self}}$.
$U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1-\chi_i)\text{rep}_b$.
if $U_{\text{self}} > U_{\text{broker}}$ **then**
    return self.
if $U_{\text{broker}} > U_{\text{self}}$ **then**
    return broker.
return self w.p. 1/2, else broker ▷ $U_{\text{self}} = U_{\text{broker}}$.

6 One Period

Local variables for PeriodUpdate

Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
α_R	scalar	Standing broker roster share.
p_{roster}	scalar	Roster-churn probability.
η	scalar	Principal exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 8 PeriodUpdate

Require: Previous state S^{t-1}, period t, parameters ϑ.
Ensure: Updated state S^t, accepted records $\mathcal{R}^t$, and pre-turnover measure row Y^t.

Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i, and set $D^t \leftarrow \emptyset$.
Refresh roster: remove each roster member with probability p_{roster}, then replenish uniformly to size $\lceil \alpha_R N \rceil$.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.
$\mathcal{D}^t \leftarrow \emptyset$.
for $i = 1, \ldots, N$ **do**
    Draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
    if $d_i^t > 0$ **then**
        $z_i^t \leftarrow \textsc{ChooseChannel}(i, \text{rep}_b^{t-1})$.
        Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
        if $z_i^t = \text{broker}$ **then**
            Add i to D^t.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$.
$\textsc{TrainPredictors}(S, t, \vartheta)$.
$\mathcal{R}^t \leftarrow \textsc{OfferMarket}(S, \mathcal{D}^t, \vartheta)$.
Synchronize broker-node edges to Roster^t, D^t, and current broker-channel matches.

UpdateSatisfaction$(S, \mathcal{D}^t, \mathcal{R}^t)$.
if $D^t \neq \emptyset$ **then**
    $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i,\text{broker}}^t$.
else
    $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
if $t \bmod \Delta_{\text{net}} = 0$ **then**
    Recompute broker betweenness, Burt constraint, and effective size on the post-matching, pre-turnover graph G^t.
$Y^t \leftarrow$ CollectMeasures$(S^t, \mathcal{R}^t, \vartheta)$ on the post-matching, pre-turnover state.
Append n_i^t to $\mathbf{p}_i^t$ for every principal and append n_b^t to $\mathbf{p}_b^t$, recording the end-of-period training-window boundaries.
for $i = 1, \ldots, N$ **do**
    if principal i exits with probability η **then**
        InitializePrincipal(S^t, i, t, ϑ).
Synchronize broker-node edges to the post-turnover roster, clients, and active matches.
return $(S^t, \mathcal{R}^t, Y^t)$.

7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \to \mathbb{R}$	Principal predictor.
$\hat{q}_b^t$	$\mathbb{R}^{d_B} \to \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i.
U_i^t	subset of $[N]$	Self-search stranger pool independently sampled for demander i.
$n_{\text{strangers}}$	integer	Per-demander stranger-pool size.
$v_{i\to j}^{\text{self},t}$	scalar	Self-offer score.
$\mathcal{U}_{b,i}$	subset of $[N]$	Candidate set for broker-client demander i.
$v_{b,i\to j}^t$	scalar	Score used to order brokered candidates for demander i.
Ω_i	integer	Residual offer quota for demander i.
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \to j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t.
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{I}_{b,i}$	list	Brokered candidates for client i, sorted by $v_{b,i\to j}^t$.
$\mathcal{B}_b^t$	list	Selected directed broker offers.
$u_{i\to j}^{\text{self}}, u_{b,i\to j}^{\text{local}}, u_{b,i\to j}^{\text{global}}$	$[0, 1]$	Independent random secondary keys used only within equal-score rankings.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and o.from.
$z^\star$	channel	Accepted-record channel.
ι	$\{0, 1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i,j) = \textsc{PredictBroker}(i,j).$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i,j) \in \mathcal{E}_G^t\}$ exclude the broker node. Principal i assesses any presented candidate as

$$a_{i\to j}^t = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i),\ C_{ij}^t > 0, \\ \textsc{PredictPrincipal}(i, x_j^t), & \text{otherwise.} \end{cases}$$

Self-search makes this score available only for neighbors and non-neighbors in the demander-specific stranger pool:

$$v_{i\to j}^{\text{self},t} = \begin{cases} a_{i\to j}^t, & j \in \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i)), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means replace predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j\leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i,j) \in \mathcal{E}_G^t,\ C_{ji}^t > 0, \\ \textsc{PredictPrincipal}(j, x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i. Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 9 AddOffer

Require: Offer book O^t, unordered-pair set $\mathcal{P}_O^t$, offer (i,j,z,v).
if O_{ij}^t is defined **then**
  return 0.
$O_{ij}^t \leftarrow (i,j,z,v)$.
if O_{ji}^t is undefined **then**
  $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i,j\}\}$.
return 1.

Algorithm 10 BrokerCandidates

Require: Broker-client demander i, state S, broker service S_b.
Ensure: Candidate set $\mathcal{U}_{b,i}$ and score function $v_{b,i\to j}^t$.
if $\mathsf{S}_b =$ full **then**
  $\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i\to j}^t \leftarrow v_b^t(i,j)$.
else if $\mathsf{S}_b =$ assessment-only **then**
  Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
  $\mathcal{U}_{b,i} \leftarrow [\mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))] \setminus (\{i\} \cup M_i^t)$ and $v_{b,i\to j}^t \leftarrow v_b^t(i,j)$.
else
  $\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i\to j}^t \leftarrow a_{i\to j}^t$. ▷ access-only
return $(\mathcal{U}_{b,i}, v_{b,i\to j}^t)$.

Algorithm 11 OfferMarket

Require: State S, demand records $\mathcal{D}^t$, parameters ϑ.
Ensure: Accepted records $\mathcal{R}^t$.

For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.
$O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
for each $(i, \text{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
    Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
    $\mathcal{U}_i^{\text{self}} \leftarrow \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))$.
    Remove from $\mathcal{U}_i^{\text{self}}$ the elements of $\{i\} \cup M_i^t$.
    Draw independent $u_{i\to j}^{\text{self}} \sim \text{Unif}(0,1)$ for each feasible j.
    $\mathcal{I}_i^{\text{self}} \leftarrow \text{argsort}_{j \in \mathcal{U}_i^{\text{self}}:\ v_{i\to j}^{\text{self},t} > r}(-v_{i\to j}^{\text{self},t}, u_{i\to j}^{\text{self}})$.
    for the first $\min(\Omega_i, |\mathcal{I}_i^{\text{self}}|)$ principals j in $\mathcal{I}_i^{\text{self}}$ **do**
        AddOffer$(O^t, \mathcal{P}_O^t, i, j, \text{self}, v_{i\to j}^{\text{self},t})$.
$\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$.
$D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t,\ \Omega_i > 0\}$.
$\mathcal{B}_b^t \leftarrow \emptyset$.
for each $i \in D_+^t$ **do**
    $(\mathcal{U}_{b,i}, v_{b,i\to j}^t) \leftarrow$ BrokerCandidates(i, S, S_b).
    Draw independent $u_{b,i\to j}^{\text{local}} \sim \text{Unif}(0,1)$ for each feasible j.
    $\mathcal{I}_{b,i} \leftarrow \text{argsort}_{j \in \mathcal{U}_{b,i}:\ v_{b,i\to j}^t > r}(-v_{b,i\to j}^t, u_{b,i\to j}^{\text{local}})$.
    for the first $\min(\Omega_i, |\mathcal{I}_{b,i}|)$ principals j in $\mathcal{I}_{b,i}$ **do**
        Append $(i, j, v_{b,i\to j}^t)$ to $\mathcal{B}_b^t$.
Draw a new independent $u_{b,i\to j}^{\text{global}} \sim \text{Unif}(0,1)$ for each $(i, j, v) \in \mathcal{B}_b^t$.
Sort $\mathcal{B}_b^t$ lexicographically by $(-v, u_{b,i\to j}^{\text{global}})$.
for each $(i, j, v) \in \mathcal{B}_b^t$ **do**
    if $\Omega_i > 0$ **then**
        $\iota \leftarrow$ AddOffer$(O^t, \mathcal{P}_O^t, i, j, \text{broker}, v)$.
        $\Omega_i \leftarrow \Omega_i - \iota$.
for each $\{i, j\} \in \mathcal{P}_O^t$ **do**
    $o_{ij} \leftarrow O_{ij}^t$ if defined, else $\emptyset$; $o_{ji} \leftarrow O_{ji}^t$ if defined, else $\emptyset$.
    if $o_{ij} \neq \emptyset$ and $o_{ji} \neq \emptyset$ **then**
        FinalizeMatch$(S, \mathcal{R}^t, o_{ij}, o_{ji})$.
    else if $o_{ij} \neq \emptyset$ and $w_{j\leftarrow i}^t > r$ **then**
        FinalizeMatch$(S, \mathcal{R}^t, o_{ij}, \emptyset)$.
    else if $o_{ji} \neq \emptyset$ and $w_{i\leftarrow j}^t > r$ **then**
        FinalizeMatch$(S, \mathcal{R}^t, o_{ji}, \emptyset)$.
return $\mathcal{R}^t$.

Algorithm 12 FinalizeMatch

Require: State S, accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o'.
if $j \in M_i^t$ **then**
    return .
Draw $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
Append columns x_j^t to Ξ_i and x_i^t to Ξ_j; append q_{ij}^t to y_i and y_j; increment n_i and n_j.
Update $\bar{q}_{ij}, C_{ij}$ and $\bar{q}_{ji}, C_{ji}$.
Add edge (i, j) to G^t; add j to M_i^t and i to M_j^t.
if $z = \text{broker}$ or ($o' \neq \emptyset$ and $o'.z = \text{broker}$) **then**
    Append the two party types x_i^t and x_j^t to $X_{b,1}$ and $X_{b,2}$; append q_{ij}^t to y_b; increment n_b.

If $\mathsf{L} = \text{Ridge}$ and $a_b = \text{single-principal}$, draw and store $r_\ell \sim \text{Unif}\{1, 2\}$ for this observation.
    $z^\star \leftarrow$ broker.
else
    $z^\star \leftarrow$ self.
Append accepted record $(i, j, z^\star, q_{ij}^t, v, o, o')$ to $\mathcal{R}^t$.

8 Satisfaction and Measures

Local variables for satisfaction

Symbol	Dimension/domain	Definition
Y_i	scalar	Principal i period satisfaction numerator.
$n_{i,b}$	integer	Accepted broker offers credited to demander i.
$\tilde{q}_i^t$	scalar	Channel-specific satisfaction input.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.

Algorithm 13 UpdateSatisfaction

Require: State S, demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.
for each $(i, z_i, d_i) \in \mathcal{D}^t$ **do**
    $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\text{from} = i\}\, \zeta.q.$
    $n_{i,b} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\text{from} = i,\ o.z = \text{broker}\}.$
    if $z_i = \text{self}$ **then**
        $\tilde{q}_i^t \leftarrow Y_i/d_i - c_s.$
    else
        $\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b})/d_i;\ \chi_i^t \leftarrow 1.$
    $s_{i,z_i}^t \leftarrow (1-\omega) s_{i,z_i}^{t-1} + \omega \tilde{q}_i^t.$

CollectMeasures. CollectMeasures$(S^t, \mathcal{R}^t, \vartheta)$ records the measures defined in Appendix B, Section 8, without changing the simulation state.